

SAGE: Steering the Adversarial Generation of Examples With Accelerations

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Abstract—To generate image adversarial examples, state-of-the-art black-box attacks usually require thousands of queries. However, massive queries will introduce additional costs and exposure risks in the real world. Towards improving the attack efficiency, we carefully design an acceleration framework SAGE for existing black-box methods, which is composed of **sLocator** (initial point optimization) and **sRudder** (search process optimization). The core idea of SAGE in terms of 1) saliency map can guide the perturbations towards the most adversarial direction and 2) exploiting bounding box (bbox) to capture those salient pixels in the black-box attack. Meanwhile, we provide a series of observations and experiments that demonstrate bbox holds *model invariance* and *process invariance*. We extensively evaluate SAGE on four state-of-the-art black-box attacks involving three popular datasets (MNIST, CIFAR10, and ImageNet). The results show that SAGE could present fundamental improvements even against robust models that use adversarial training. Specifically, SAGE could reduce >20% of queries and improve the success rate of attacks to 95%~100%. Compared with the other

acceleration framework, SAGE fulfills the more significant effect in a flexible, stable, and low-overhead manner. Moreover, our practical evaluation (Google Cloud Vision API) shows SAGE can be applied to real-world scenarios.

Index Terms—Adversarial examples, saliency map, bounding box, black-box attacks, acceleration framework.

I. INTRODUCTION

DEEP Neural Networks (DNNs) are becoming ubiquitous in security-critical applications to deliver automated decisions such as face recognition, self-driving cars, malware detection, etc [47], [48], [50]. Subsequently, several security concerns have emerged regarding the potential vulnerabilities of the DNN algorithm itself [2]. Particularly, adversaries can deliberately craft special inputs, named *adversarial examples* (AEs), leading models to produce an output for their malicious intentions, such as misclassification.

In recent years, the research and industry communities have proposed a variety of attack approaches on image classifiers [4], [49]. Depending on the capabilities of attackers, prior works can be divided into white-box and black-box attacks. In white-box attacks, the adversary can get the gradient information of the model inference [8], [26], [34]. Even some adversaries have full knowledge of the network, including the architecture and all parameters [8], [16], [33], [34], [42]. In black-box attacks, the adversary faces more challenges [1], [19], [45] as it has limited model information when generating AEs. Furthermore, black-box attacks can be categorized into three types, notably, *score-based attack* [10], [20], [41], *transfer attack* [11], [29], [36], [44], and *decision-based attack* [5], [6], [9], [17], [43].

Among them, score-based attacks [20], [45] estimate the gradient by querying the prediction confidence from the target model. Decision-based attacks [6], [9] solely rely on the predicted labels of the DNN model. They use heuristic search algorithms to iterate along the model decision boundary. Transfer attack is based on the transferability of AEs between different model structures [44]. In this method, previous work assumes the attacker can train a substitute model or access similar training data used for the victim model [41].

It is not easy to obtain the model information required by white-box attacks, so black-box attacks are more practical in real scenarios. However, the existing black-box attacks are query-inefficient since they have only limited model prediction results. Specifically, the decision-based [6], [9] and score-based [20], [45] attacks usually require a large

Manuscript received 21 March 2022; revised 8 November 2022; accepted 9 November 2022. Date of publication 2 February 2023; date of current version 8 February 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62227805 and Grant 62072398, in part by the National Key Research and Development Program of China under Grant 2020AAA0107700, in part by the Alibaba-Zhejiang University Joint Institute of Frontier Technologies, in part by the Zhejiang Key Research and Development Plan under Grant 2021C01116, in part by the Leading Innovative and Entrepreneur Team Introduction Program of Zhejiang under Grant 2018R01005, in part by the Research Institute of Cyberspace Governance in Zhejiang University, in part by the National Key Laboratory of Science and Technology on Information System Security under Grant 6142111210301, in part by the State Key Laboratory of Mathematical Engineering and Advanced Computing, and in part by the Open Foundation of Henan Key Laboratory of Cyberspace Situation Awareness under Grant HNTS2022001. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Stefano Tomasini. (Corresponding author: Fan Zhang.)

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Digital Object Identifier 10.1109/TIFS.2022.3226572

number of queries to achieve heuristic search and estimate the gradient direction, respectively. These massive queries might incur additional service charges [46] when attacking machine-learning-as-a-service (MLaaS) platforms. For instance, to attack the Google Cloud Vision API [12], the Hybrid attack [41] or HSJA attack [9] takes about 15,000 queries to successfully generate a targeted adversarial example in the ImageNet dataset, which means costing \$22.5 for the desired AE. Moreover, large-number queries also increase the exposed risk of the adversary.

It is actually in demand to put efficient black-box attacks into practice when generating AEs or evaluating the victim model robustness. Toward this end, we need to understand the intrinsic mechanism of the black-box attack and optimize its searching process. According to prior literature [37], the saliency map of an input sample contains the most useful information for the model prediction. In other words, the adversary could use the saliency map to guide the attack direction. However, the saliency map cannot be calculated directly due to lacking gradient information in the black-box attack. To address this problem, we propose to *exploit the bounding box (bbox) to capture the saliency map pixels*. Intuitively, perturbing those pixels in the bbox is assumed to contribute the most to misclassification. In addition, we provide a series of observations to support the universality in common models and stability during the attack process, named *model invariance* and *process invariance* respectively. Center around the above idea, we design an acceleration framework SAGE (Steering the Adversarial Generation of Examples, for short) for the black-box attack. SAGE consists of two components, *i.e.*, **sLocator** and **sRudder**.

(i) **sLocator** optimizes the initial step of a black-box attack by finding a promising starting point that is close to the successful adversarial example. Specifically, it fuses the salient features of the target sample with the original input sample to get an initial point. This means that **sLocator** can accurately obtain the semantic information of the target class while introducing a limited distance. Therefore, **sLocator** realizes to jump to a favorable position while minimizing the number of unnecessary queries on non-salient areas.

(ii) **sRudder** accelerates the remaining searching process by guiding the perturbation. The previous methods [6], [9], [19], [41], [45] treat all pixels equally and perturb the whole image. Yet **sRudder** uses bbox to adjust the perturbation in each iteration, thus it can benefit to accurately attack the salient area of the image. **sRudder** sits in SAGE following an incremental and gradual manner, just like the rudder navigating the vessel towards the destination quickly and safely.

Contributions: Our contributions are summarized following:

- We explore the intrinsic relationship between the saliency map and the bounding box, as well as provide two observations including model invariance and process invariance. Subsequently, we advance the core idea is using bbox to capture saliency map pixels in the black-box attack.
- Pursuing the core idea, we propose SAGE, a universal framework designed specifically to serve as an accelerator for black-box attacks. It exploits the saliency area in the

bbox to improve starting point and search process, hence achieving the query-efficient attack.

- We perform extensive evaluations to demonstrate the effectiveness of SAGE: (i) Accelerate the state-of-the-art (SOTA) methods from score-based attacks and decision-based attacks. (ii) Compare SAGE with the Hybrid attack framework (combination of score-based attack and transfer attack) in terms of acceleration effect, robust model testing, and time overhead. (iii) Enforce practical evaluations on Google Cloud Vision API [12] to test the attack effect in the real world. (iv) Evaluate the stability of SAGE against diverse bbox size and imprecise bbox.

II. RELATED WORK

A. White-Box Attack

In the white-box environment, most of the current methods [8], [16], [33], [34], [42] generate AEs based on the model information such as its internal architecture and gradient. For example, the Fast Gradient Sign Method (FGSM) [16] and Projected Gradient Descent (PGD) [33] produce perturbation for maximizing the inner part of the saddle point formulation. The white-box attack can achieve a high success rate with some queries. Yet white-box attack requires inherent information in the model, which limits its application scenarios in the real world.

B. Black-Box Attack

Compared to white-box settings, black-box attacks are more applicable in practice. The previous black-box methods mainly include the score-based attack, transfer attack, and decision-based attack. In score-based attacks, Chen et al. [10] realize the AEs generation by comparing the finite differences pixel by pixel. Ilyas et al. [20] propose the Bandits_{TD} attack by integrating time and data-related information based on the NES attack [19]. In transfer attacks [11], [29], researchers suggest improving the black-box attack with the transferability of AEs. There is also a hybrid framework [41] suggested by combining score-based and transfer attacks. However, score-based and transfer attacks [25], [36] are prone to failure when the victim model uses adversarial training by introducing controllable noise or defensive distillation to reduce transferability. For the decision-based attack, Brendel et al. [5] and Brunner et al. [6] propose to perform heuristic AE searches along the decision boundary. Those approaches are not efficient enough and usually require massive queries. In § III-B, we will further outline these black-box attacks to lead out our design inspiration.

III. PRELIMINARY

A. Threat Model and Assumptions

We first define here the notations used in this paper and introduce the threat model. Consider a DNN image classifier that achieves to learn a model $f: \mathcal{X} \rightarrow \mathcal{Y}$, where $x \in \mathcal{X} \in \mathbb{R}^d$ is the input, and $\mathcal{Y} = \{1, 2, \dots, L\}$ with L being the number of classes. Let y denote the ground-truth label of x , and x_{adv} denotes an AE for x .

Adversary's Goals: An adversary could perform untargeted or targeted attacks. The untargeted attack aims to cause the classifier misclassification, i.e., $f(x_{adv}) \neq y$. The targeted attack refers to fooling the model into misclassifying the input sample as the adversary-desired class, i.e., $f(x_{adv}) = y^*$, where y^* is the target class. Formally, the attack aims to find an example x_{adv} to meet the following conditions

$$f(x_{adv}) \neq y \quad \text{s.t.} \quad \|x_{adv} - x\|_p \leq \varepsilon \quad (1)$$

where $\|\cdot\|_p$ denotes the ℓ_p distance and ε is the acceptable perturbation threshold (e.g., imperceptible to human eyes). We consider ℓ_2 and ℓ_∞ in this paper.

Assumptions: We assume that the trained DNN model can correctly identify most of the samples. In other words, the trained model can recognize image semantics to achieve multi-classification well in the actual scenario.¹ Meanwhile, in the black-box attack, the attacker has no knowledge of the training set but can collect some data containing several samples of target classes. This is the same assumption as plenty of literature about black-box attacks [13], [27], [28], [31], [38], [41]. Both trained parameters and model architectures are unknown.

B. Black-Box Adversarial Attack Overview

In this section, we outline the current work in terms of black-box attack and acceleration framework.

1) *General Pipeline of Black-Box Adversarial Attack:* In the landscape of black-box attacks, score-based and decision-based methods² present a general pipeline for AE generation, including step 1: locating the initial point, and step 2: the search process.

a) *Score-based attack:* The score-based attack uses the predicted confidence information to estimate the gradient direction of the target model. For example, Ilyas et al. [19] propose a new attack method NES based on random vector gradient estimation. Tu et al. [45] design AutoZOOM that is based on the two-point estimation of random vectors, and it reduces the number of required queries for each Carlini-Wagner (CW) iteration without losing too much accuracy. These methods require many queries to estimate the gradient and face a risk of failure when the victim adopts robust models.

b) *Decision-based attack:* Compared with the above methods, the decision-based attack solely needs the predicted label (the class with maximum probability). To generate AEs, it initializes with a target-class sample and perturbs the image along the decision boundary. For example, HSJA [9] uses Monte Carlo to estimate the gradient direction to implement a decision-based attack. Biased boundary attack [6] utilizes biased methods (such as low-frequency perturbation) to improve the search process. However, they are inefficient given that requiring a large number of queries to perform heuristic searches.

¹Particularly, Ribeiro et al. [39] mention a classifier that has learned to differentiate wolves from huskies based on image background. But this model is not universally applicable to multi-classification problems due to lacking robustness.

²Transfer attack mainly exploits the transferability of AE between different models, so we outline score-based and decision-based attacks here. Furthermore, the transfer attack will be a part of the Hybrid attack framework [41].

2) *Hybrid Attack Framework:* The Hybrid attack [41] is a framework that combines score-based attack and transfer attack. Where the transfer attack is based on the transferability of AEs between different model structures [44]. Hybrid attack first leverages the PGD [33] to generate an attack seed and uses it to perform a score-based attack. Then it applies the generated AEs to fine-tune the local DNN model. Especially, the Hybrid attack optimizes AutoZOOM and NES based on the transfer attack and has considerable improvement. However, this proposal will be limited when PGD or transfer attack fails (against robust models). Moreover, the Hybrid attack framework is not flexible enough since it requires fine-tuning the local model.

C. Saliency Map

We introduce the background of saliency map [37] which we use to guide the search process during the attack. We first clarify the concept of forward derivative before defining the saliency map. As the basis of deep neural networks, the forward derivative is the Jacobian matrix of the network F , and it can be learned during training. Papernot et al. [37] state that the Jacobian matrix $J_F(X)$ can be successively computed from the input layer X to the output layer. Specifically, for a DNN model with N categories of output, the forward derivative for the given M -dimensional sample X as following:

$$J_F(X) = \frac{\partial F(X)}{\partial X} = \left[\frac{\partial F_j(X)}{\partial x_i} \right]_{i \in [1, M], j \in [1, N]} \quad (2)$$

Adversarial saliency maps are defined to suit problem-specific goals. Considering a DNN classifier, we use $F_j(X)$ to denote the probability of inferring the sample X as j . The final predicted result corresponds to the component with the highest probability: $\text{label}(X) = \arg \max_j F_j(X)$. We set the saliency map based on the forward derivative, which provides the adversary with the required information to cause the neural network misclassification. For instance, suppose the adversary wants the model to misclassify a sample X as the target class $t \neq \text{label}(X)$. To this end, the target class probability $F_t(X)$ assigned by F must increase while the probabilities $F_o(X)$ of all other classes $o \neq t$ decrease, until $t = \arg \max_j F_j(X)$. To fulfill this goal, the adversary can perturb the input image as following saliency map $S(X, t)$:

$$S(X, t)[i] = \begin{cases} 0 & \text{if } J_{it}(X) < 0 \text{ or } \sum_{j \neq t} J_{ij}(X) > 0 \\ J_{it}(X) \left| \sum_{j \neq t} J_{ij}(X) \right| & \text{otherwise} \end{cases} \quad (3)$$

where i is a pixel of the input image, and $J_{ij}(X)$ denotes $J_F[i, j](X) = \frac{\partial F_j(X)}{\partial x_i}$. The specified condition on the first line in Eq. (3) rejects input components with a negative target derivative or an overall positive derivative on all other classes. Indeed, $J_{it}(X)$ should be positive for $F_t(X)$ to increase when feature X_i increases. Similarly, $\sum_{j \neq t} J_{ij}(X)$ needs to be negative to decrease or stay constant when the feature X_i is increased. The product on the second line in Eq. (3) allows us to consider all other forward derivative components together,

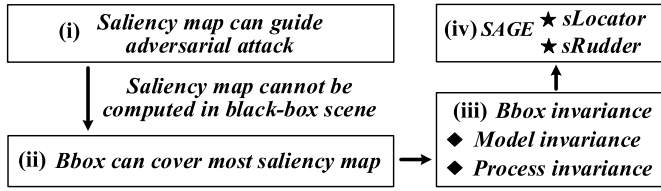


Fig. 1. High-level idea of SAGE.

so that we can compare $S(X, t)[i]$ for all input features. More mathematical justification is detailed in [37]. Overall, high values of $S(X, t)[i]$ correspond to input features that will either increase the target class, or decrease other classes significantly, or both. By increasing these input pixels, the adversary eventually makes the model misclassify the sample into target class t . This means that perturbing pixels in the saliency map area are more efficient in fooling the DNN model into misclassification.

IV. ARCHITECTURE AND OVERVIEW

In this section, we elaborate on the design principles and overview of SAGE.

A. Design Principle

We depict SAGE's design principles in Fig. 1, which contains the inspiration and four observations on accelerating the black-box attack.

Observation 1 (Saliency Map Can Guide Adversarial Attack): As we stated in § III-C, adversarial saliency maps are defined to suit problem-specific goals. Since the value of the saliency map is calculated according to the forward propagation, changing the saliency pixels can indeed achieve the adversarial goal. Thus, it is widely applicable for DNN classifiers, as long as they utilize forward propagation to perform prediction.

Observation 2 (Bbox Can Cover Most Salient Pixels): Although the saliency map can be used to accelerate the adversarial attack, it cannot be directly calculated in the black-box scenario due to the lack of forward propagation information. The bounding box refers to the rectangular box that just covers the classification object. As we stated in § III-A, SAGE focuses more on common models that can recognize object semantics in real-world scenarios. Therefore, bbox can cover most of the saliency map pixels for correctly predicted samples.

Observation 3 (Bbox Invariance): Before using bbox to capture saliency maps, we need to consider two problems.

(i) *Model invariance.* Whether is highly coincident between bboxes of different models based on the same image. We do not intend to locate the bbox of every possible model because the model architecture is unknowable in a black-box scenario. Therefore, we use a general object detection model M_{bbox} (e.g., Libra R-CNN [35]) to predict bbox to capture the saliency pixels of existing common models. We call this universality in common models as *model invariance*.

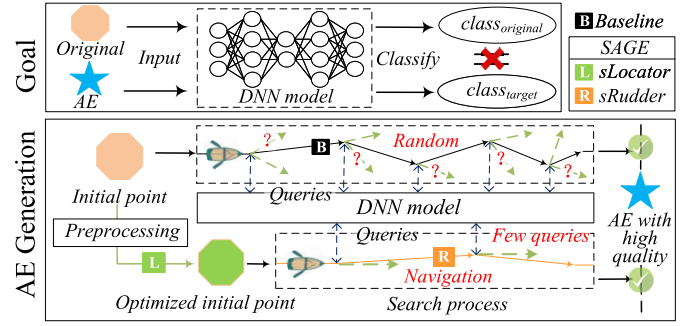


Fig. 2. Illustrative explanation of SAGE Framework.

(ii) *Process invariance.* In fact, the input sample will be perturbed and changed constantly during the process of adversarial attacks. Thus, we need to consider whether the bbox of the initial sample can cover the most salient pixels during the attack process because the saliency map changes with the input. This consideration is called *process invariance*.

Intuitively, model invariance and process invariance can be both confirmed, and we will evaluate them in the experiment (§ VI-B).

Observation 4 (Accelerate the Black-Box Attack With SAGE): Once the bbox invariances are determined, we can exploit the bbox to capture the saliency map and further accelerate the attack. Next, we elaborate on the overview of SAGE, the design-specific acceleration framework for black-box attacks.

B. SAGE Overview

The main goal of SAGE is to reduce the number of queries when generating a high-quality adversarial example. We design two components (sLocator and sRudder) that utilize bbox to capture salient pixels to advance the black-box attacks. As Fig. 2 shows, sLocator and sRudder accelerate the two steps of a black-box attack, respectively. (i) In the initialization step, sLocator can find a starting point that is close to the final AE; (ii) In the searching step, sRudder can steer the perturbation direction in each query based on the salient features. Note that we set the same success condition for SAGE and baseline methods, which means SAGE can achieve query-efficient attacks while guaranteeing the generation of high-quality AEs.

V. DESIGN DETAILS OF SAGE

In this section, we present SAGE, an acceleration framework for black-box adversarial attacks. The main idea is to use the image bbox to capture the salient features, which contribute the most to the prediction results of the DNN classifier. Guiding the attack to perturb these features could more efficiently fool the model into misclassification. Algorithm 1 depicts the acceleration framework pipeline at a very high level. When performing an attack for the seed image, we first extract the bbox in preprocessing. Then sLocator transfers the starting point to a favorable position, while sRudder will navigate the perturbation direction in every search. We also

Algorithm 1 Attack Acceleration Framework**Input:** Set of seed image X ; DNN model f ; target class t **Output:** Set of successful AEs E

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1:  $S \leftarrow X$  (remaining seeds to attack)
2:  $E \leftarrow \emptyset$  (successful AEs)
3: while  $S$  is not empty do
4:   Select the seed image to be attacked
5:    $x \leftarrow \text{select\_seed}(S)$ ;  $\text{remove}(S, x)$ 
6:   Select randomly a target image
7:    $T \leftarrow \text{select\_target}(t)$ 
8:   Preprocessing: extract the bbox
9:   (i) Step 1: initial point
10:   $x^* \leftarrow \text{new\_point}(T, x, \text{sLocator})$ 
11:  (ii) Step 2: search process
12:   $x^* \leftarrow \text{blackBoxAttack}(x^*, x, f, \text{sRudder})$ 
13:  if  $\text{is\_success}(x^*, x, f)$  then
14:     $\text{insert}(E, (x, x^*))$ 
15:  end if
16: end while
17: return  $E$ 

```

design the expansion mechanism and attenuation mechanism for **sLocator** and **sRudder** to offset the adverse effect of imprecise bbox. In subsequent subsections, we elaborate on preprocessing and two key components: **sLocator** and **sRudder**.

A. Preprocessing: Extracting Bbox

It is challenging to calculate the saliency map in the black-box setting since lacking gradient information. To tackle this task, we propose to extract bbox to capture the saliency map. As stated in § III-A, we can get some instances with the ground-truth labels. We use this dataset to train a bbox extractor. Considering the problem of data imbalance due to random selection and limited samples, we adopt Libra R-CNN [35], an imbalance-friendly model architecture. Fig. 3 shows the model architecture for bbox extraction, the IoU-balanced sampling, balanced feature pyramid, and balanced L1 loss are designed to mitigate the imbalance of sample, feature, and objective level. For example, the sample-level strategy refers to focusing on hard samples since they are effective to improve detection performance. And integrating features from each resolution to possess feature balance, as well as adjusting the loss function on classification and localization to offset the objective imbalance. More model details can be found in Libra R-CNN [35]. Note that the bbox extractor can be replaced and we are not against other SOTA models in this landscape. Then the extracted bbox can be used by **sLocator** and **sRudder** respectively to optimize the starting point of the attack and the search process.

B. sLocator: Jumping to a Favorable Position

In the black-box attack, consider the target class is t and x_A is a randomly selected sample that can be identified as t by the model. **sLocator** works in four steps. (i) Extract the bbox x_{bbox} of x_A in preprocessing. (ii) Set the coverage

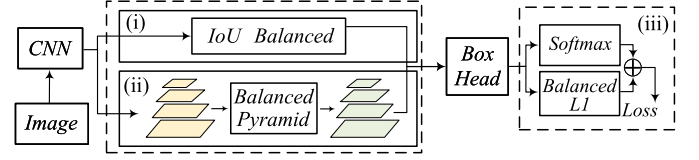
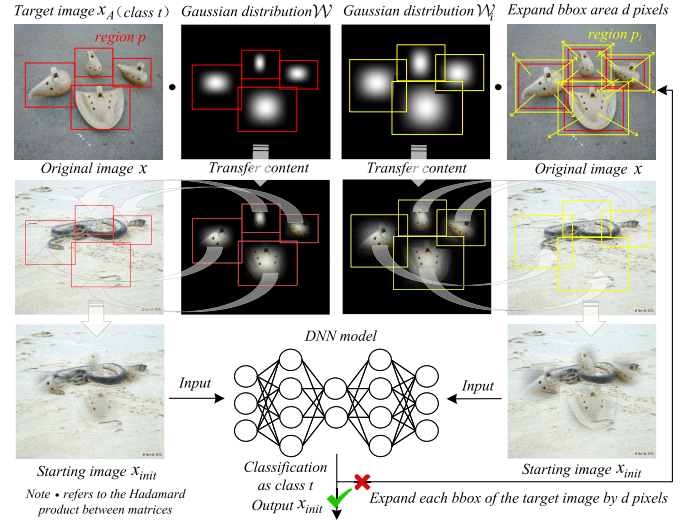


Fig. 3. Architecture of Libra R-CNN for extracting bbox.

Fig. 4. Process of **sLocator**.

weight matrix $\mathcal{W}$. (iii) For the x_{bbox} area of original image x , cover the pixels with the weighted target image x_A , i.e., $x_{init} = (x_A - x) \cdot \mathcal{W} + x$, where “ $\cdot$ ” represents the Hadamard product (i.e., element-wise multiplication) between matrices. To accelerate score-based methods, x_{init} can be directly used as a new starting point for the attack. Fig. 4 shows how **sLocator** uses the weight matrix $\mathcal{W}$ with the two-dimensional Gaussian distribution³ to generate the initial point. Assuming that the size of the bbox region is $a \times b$, this two-dimensional Gaussian distribution can be regarded as a product of two Gaussian distributions with zero covariance and equal variances, expressed as:

$$f(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{(x-\mu)^2 + (y-\mu)^2}{2\sigma^2}}, \quad -\frac{a}{2} \leq x \leq \frac{a}{2}, \\ -\frac{b}{2} \leq y \leq \frac{b}{2}. \quad (4)$$

(iv) Particularly, the targeted decision-based attack requires that the starting point of the attack must be identified as the target class by the DNN model. To this end, **sLocator** presents an expansion mechanism. Fig. 4 provides an illustrative explanation for this process. When x_{init} cannot be recognized as the target class, the bbox region p is expanded by d pixels. Iterating this regional growth until the picture is recognized as the desired class. In the worst case, region p will expand

³The consideration of using Gaussian distribution mainly stems from two aspects. For one thing, Gaussian distribution, in a gradual manner, causes fewer pixel mutations and introduces less ℓ_p distance than overall area coverage. For another, Gaussian noise is generally applicable to adversarial perturbations [15], [40] given its effectiveness and imperceptibility [30].

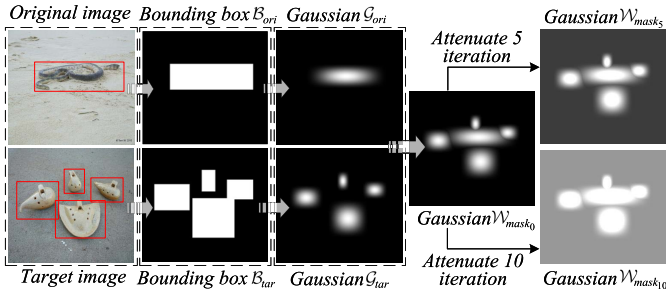


Fig. 5. Generation process of Gaussian mask in sRudder.

to the whole image, *i.e.*, the starting point is the target image, achieving the same effect as baselines.

Moreover, the expansion mechanism can offset the adverse impacts of imprecise bbox. For instance, when the bbox has an accuracy loss that cannot contain most of the salient features, the starting point x_{init} cannot be predicted as the target class. Through the expansion mechanism, the neglected saliency pixels due to the inaccurate bbox will gradually be considered. Specifically, we provide some evaluations in § VI-I to analyze the performance of SAGE with an imprecise bbox.

C. sRudder: Navigating the Perturbation Direction

During the searching process of a black-box attack, the existing approaches will generate a set of random noise for estimating the gradient or performing the heuristic search. In contrast to the aimless perturbation on the whole image, sRudder uses a mask to concentrate the perturbation matrix on the saliency map. The mask will be initialized as a Gaussian matrix⁴ and gradually decay during the attack process.

1) *Gaussian Mask Generation*: We explain how to generate a Gaussian mask in Fig. 5. (i) Obtain bboxes $\mathcal{B}_{ori}$ and $\mathcal{B}_{tar}$ of the original image and the target image respectively. Particularly, if the expansion mechanism is triggered in sLocator, the $\mathcal{B}_{tar}$ refers to the bbox region after expanding, *i.e.*, the bbox of the target image used in sLocator and sRudder is consistent. (ii) Use $\mathcal{B}_{ori}$ and $\mathcal{B}_{tar}$ to generate the corresponding Gaussian matrices $\mathcal{G}_{ori}$ and $\mathcal{G}_{tar}$. (iii) The Gaussian mask $\mathcal{W}_{mask}$ is calculated as the union of $\mathcal{G}_{ori}$ and $\mathcal{G}_{tar}$.

2) *Attenuation Mechanism*: Then we set the attenuation mechanism to adapt to the dynamic attack process. Specifically, given the number of attenuation iterations N , attenuation coefficient α , and maximum weight β . For the initial mask $(\mathcal{W}_{mask})_0$; set $w_j \in (\mathcal{W}_{mask})_i$, corresponding position $w'_j \in (\mathcal{W}_{mask})_{i-1}$ ($i \geq 1$), where $(\mathcal{W}_{mask})_i$ is used in the i -th perturbation, we attenuate the mask as follows:

$$w_j = \begin{cases} w'_j + \alpha / \sqrt{i+1} & \text{for } w'_j \leq \beta - \alpha / \sqrt{i+1}, \\ \beta & \text{otherwise.} \end{cases} \quad (5)$$

Fig. 5 shows $(\mathcal{W}_{mask})_5$ and $(\mathcal{W}_{mask})_{10}$ when $\alpha = 0.1$, and $\beta = 1$. After N -round perturbation, all values will be set as 1, that is, $\forall w_j \in (\mathcal{W}_{mask})_{N+1}: w_j = 1$.

⁴To maintain consistency between sLocator and sRudder, a Gaussian mask is applied. The insight behind is that the mask assigns a larger weight to the regions significantly covered by sLocator and vice versa. Meanwhile, the Gaussian mask could be beneficial to focus on the attack target position and weak the edges [7].

3) *Attack With sRudder*: After generating Gaussian weight $\mathcal{W}_{mask}$, every perturbation will be guided by sRudder during the attack. sRudder adjusts each random noise (denoted as $diff_i$) used for boundary search or gradient estimation to focus on the $\mathcal{W}_{mask}$ area, *i.e.*, $diff_i^* = diff_i \cdot \mathcal{W}_{mask}$. Therefore, $diff_i^*$ is the perturbation matrix used in SAGE. As the number of iterations increases, $\mathcal{W}_{mask}$ will decay until it becomes the all-one matrix. The attenuation mechanism can benefit the fine-tuned perturbation in the later stage of attacks (even with imprecise bbox).

D. An Attack Instance With SAGE

We present an attack instance with sLocator and sRudder in Fig. 6. It displays a targeted attack that attacks the desktop computer into a lampshade. The bottom part shows the attack process of HSJA [9], and the top part presents the accelerated process with SAGE. We analyze two steps of the attack process.

(i) Step 1: HSJA directly uses the target picture as the starting point of the attack. While in SAGE, sLocator covers the bbox area of the target image on the original image. Perform the expansion mechanism until the covered picture can be identified as the target class, thereby obtaining a new starting point. (ii) Step 2: HSJA exploits Monte Carlo to estimate the gradient to determine the perturbation of each iteration. For the search step in SAGE, sRudder leverages the delicate-design mask matrix to adjust each random noise to focus on the $\mathcal{W}_{mask}$ area. As Fig. 6 shows, the perturbation guided by sRudder is mainly concentrated in the bbox area, instead of evenly perturbing the entire picture, just like the baseline method. With the attenuation of $\mathcal{W}_{mask}$, the perturbed area expands and the ℓ_p distance is gradually reduced. Finally, HSJA requires 12,498 queries to generate an AE with $dist_2 = 20.14$. While SAGE uses 9,341 queries and generates an AE with $dist_1 = 10.33$.

E. Deep Insights of SAGE

To further understand SAGE, we visualize the saliency map to provide deep insights. Particularly, we select and analyze four SOTA black-box attacks: HSJA (decision-based), NES (score-based), AutoZOOM (score-based), and the Hybrid Attack (combination of score-based attack and transfer attack).

Fig. 7 shows a targeted attack that attacks a ruler into a plastic bag. Subfigures (h)-(m) display the perturbation of SAGE and other SOTA black-box attacks. Red presents the perturbation on the saliency map, while blue corresponds to the non-saliency map. It is clear that the perturbations of SAGE focus on the saliency map area (*i.e.*, $\sim 92.3\%$), while the attack areas in prior methods are distributed across the whole image.

VI. EVALUATION

In this section, we comprehensively evaluate SAGE in terms of the number of queries (QUE), success rate (SUC), and time overhead. All evaluations are performed on the GTX 1070 GPU, with code available online.⁵ Our experiments are designed to answer the following research questions:

⁵See repository <https://github.com/Secbrain/SAGE>

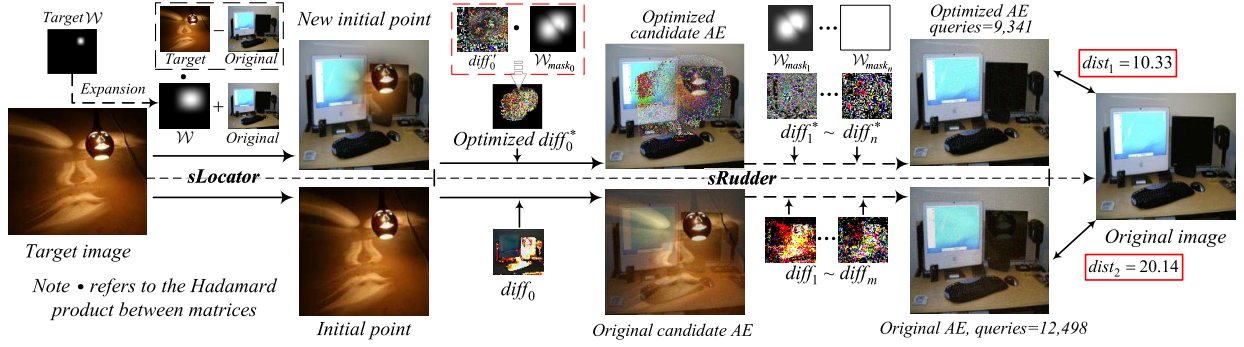


Fig. 6. Targeted attack instance with SAGE. It attacks the desktop computer into a lampshade. Bottom part: HSJA [9]; Top part: SAGE.

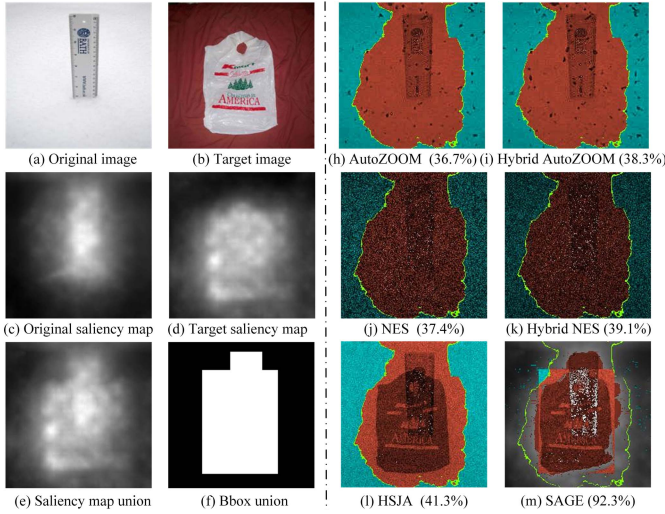


Fig. 7. Perturbation visualization. Subfigures (a)-(f) show the saliency map and bbox. And subfigures (h)-(m) display the perturbation of each method. Red indicates the disturbance on the saliency map, while blue corresponds to the non-saliency map. The digital (area ratio): red/red+blue.

RQ1. Is bbox invariance universal in common models and during the attack process?

RQ2. How about the acceleration effect of SAGE (including integration tests and ablation experiments) for decision-based and score-based attacks?

RQ3. Compared with the Hybrid attack framework, what are the advantages of SAGE?

RQ4. Can SAGE be used in practice?

RQ5. How does SAGE perform when different target images are selected?

RQ6. How does the acceleration effect differ across various bbox sizes and numbers?

RQ7. How stable is SAGE based on imprecise bbox?

A. Experiments Setup

1) Baselines: We compare SAGE with four state-of-the-art black-box attacks: HSJA [9] (decision-based attack), AutoZOOM [10] (score-based), NES [20] (score-based), and the Hybrid attack framework [41]. All the implementation and hyper-parameters of baselines are from the public source code.

2) Datasets and Models: We perform evaluations on common models and three popular image classification datasets, *i.e.*, MNIST, CIFAR10 [24], and ImageNet [22]. Among them, the MNIST contains 70K 28×28 grayscale images of handwritten digits from 0 to 9. The CIFAR10 [24] includes 60K 32×32 RGB images, divided into 10 classes. The ILSVRC2012-version verification set of the ImageNet contains 50K real-world RGB images (1,000 classes). In § VI-C, § VI-D, and § VI-E, we randomly selected 1,000 images from ImageNet as the test set. For MNIST and CIFAR10, we randomly selected 100 images from each class ($10 \times 100 = 1,000$) to evaluate. These settings are consistent with baseline methods.

For the victim DNN models, our settings are the same as the baseline. Specifically, comparing HSJA, we choose LeNet, 20-layer ResNet, and 50-layer ResNet for MNIST, CIFAR10, and ImageNet, respectively. Comparing AutoZoom, NES, and the Hybrid attack, we use a series of normal and robust models. For MNIST, we use the model trained by Bhagoji et al. [3] and Madry’s MNIST robust model [33]; For CIFAR10, we use the DenseNet model trained by Fnu et al. [41] and Madry’s CIFAR10 robust model [33]; For ImageNet, we use the DenseNet [18]. Note that the above robust models can obfuscate the gradient information or prevent transfer attacks through controllable noise and defensive distillation mechanisms [33]. This means that it is more difficult to estimate the gradient, and the transfer attack is also limited.

3) Initialization: Both original classes (and target classes in targeted attacks) and their samples are randomly selected. Moreover, we randomly select 10% of each dataset for training Libra R-CNN to extract bbox. Meanwhile, the hyperparameters of SAGE need to be initialized before performing attacks. Specifically, the per-iteration expansion size d is set 5 in sLocator. In sRudder, we set the number of attenuation iterations $N = 100$, attenuation coefficient $\alpha = 0.05$, and maximum weight $\beta = 2$, respectively. If not otherwise stated, all results are averaged over 5 runs based on different random seeds (consistent with previous work [41]).

4) Metrics: On the one hand, we compute the success rate (SUC for short) of the attack in a pre-set condition. On the other hand, we analyze the number of queries (QUE for short) when the ℓ_p distance is less than a certain threshold.

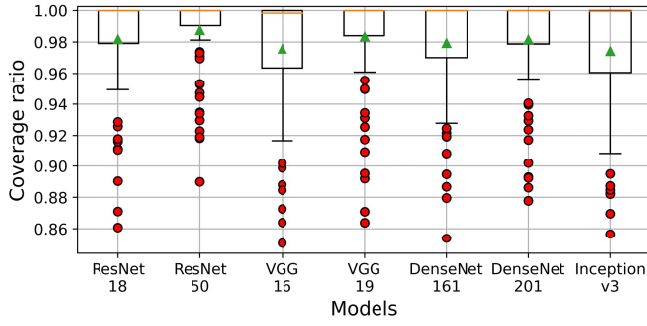


Fig. 8. Saliency map coverage ratio in bbox.

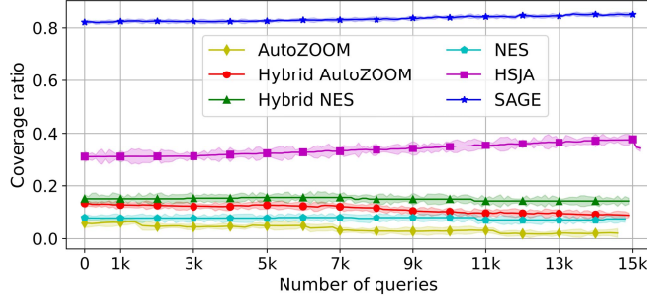


Fig. 9. Proportion of perturbation in saliency map for black-box attacks.

Particularly, the ℓ_2 distance refers to the Euclidean distance, while ℓ_∞ distance represents the maximum absolute value of the perturbation matrix. Furthermore, we also compare SAGE with the Hybrid attack in terms of acceleration effect and introduced time overhead.

B. Experiment 1: Evaluation on Bbox Invariance (RQ1)

The evaluations include model and process invariance.

(i) Model invariance. As the observation 3 stated in Fig. 1, invariant representation of bbox between different models supports that SAGE is universally applicable against diverse models. We exploit Libra R-CNN [35] as the preprocessing model to extract bbox, and evaluate the saliency map coverage in 7 common models, including ResNet-50, ResNet-18, VGG-16, VGG-19, DenseNet-161, DenseNet-201, and Inception-V3 (from <https://keras.io/applications/>).

Subsequently, we randomly select 5,000 images from the ImageNet dataset. For each image, the notation Sal_{bbox} represents the number of saliency map pixels contained in the extracted bbox (by Libra R-CNN), and Sal_{all} denotes the number of pixels in the saliency map (of each model). Then we calculate the proportion of saliency map pixels falling on the bbox, that is: $Cov_{ratio} = Sal_{bbox} / Sal_{all}$. As shown in Fig. 8, the bbox can contain more than 90% of the saliency map area in all models. On average, about 98% of the saliency map pixels fall in the bbox. Therefore, using the pre-extracted bbox to capture the salient features is universally applicable for common models.

(ii) Process invariance. The second evaluation part of bbox is process invariance. It is also important to ensure bbox can always cover most of the saliency map during the attack processing because the saliency map changes with the current

TABLE I
EVALUATE SAGE ON HSJA

ℓ_p	Data	Model	Objective	Model Queries			
				HSJA	L	R	L+R
ℓ_2	MNIST	LeNet	Untargeted	1,359	1,204	1,171	1,098
			Targeted	12,365	11,665	10,356	9,399
	CIFAR10	ResNet	Untargeted	1,030	858	998	758
			Targeted	5,519	4,545	4,870	4,901
	ImageNet	ResNet	Untargeted	976	894	859	805
			Targeted	10,354	9,458	9,018	8,272
ℓ_∞	MNIST	LeNet	Untargeted	3,146	2,986	2,705	2,897
			Targeted	4,480	2,960	3,849	3,405
	CIFAR10	ResNet	Untargeted	1,978	1,900	1,834	1,600
			Targeted	3,517	3,381	3,116	2,961
	ImageNet	ResNet	Untargeted	13,245	11,287	11,951	10,978
			Targeted	15,080	13,658	13,934	12,214

input sample. We perform 15,000 queries for 5,000 attack instances. Fig. 9 shows the ratio of the saliency map in the perturbation area during SAGE and those SOTA black-box attacks. The coverage ratio has only slight fluctuations in all methods with the changing input during the attacks, as well as the deviation of the coverage ratio is relatively small (especially in SAGE). This phenomenon confirms the process invariance of bbox.

Moreover, for the previous methods, only 10%~40% of the perturbation area is on the saliency map in each iteration. This means their attacks perturb many ineffective pixels, which leads to large-number queries. While the perturbations of SAGE cover >80% saliency map points throughout the attack process.

Remark 1: Using bbox can cover most of the saliency pixels for common DNN models, i.e., model invariance. During the attack process, the proportion of saliency pixels in each type-attack perturbation tends to be stable, i.e., process invariance.

C. Experiment 2: Acceleration on Decision-Based Attack (RQ2)

HSJA [9], a decision-based attack based on Monte Carlo to estimate gradient direction along the decision boundary. To evaluate SAGE on HSJA, we adopt the same experimental setting as the baseline to measure the optimization performance. Specifically, the ℓ_2 thresholds of datasets for targeted attacks and untargeted attacks are MNIST (1.48, 0.8), CIFAR10 (0.5, 0.3), and ImageNet (25.89, 15.89), respectively. The corresponding ℓ_∞ thresholds are MNIST (0.175, 0.03), CIFAR10 (0.05, 0.01), and ImageNet (0.5, 0.15) in turn. Table I summarizes the average query number when HSJA and SAGE (L: sLocator, R: sRudder) attack successfully. To obtain the adversarial examples with the same quality, SAGE requires significantly fewer queries than HSJA. For example, in the ℓ_2 attacks of MNIST, SAGE accelerates average queries from 1,359 to 1,098 on untargeted (reduced 19.2%) and from 12,365 to 9,399 on targeted (reduced 24%). Similarly, SAGE achieves over 20% acceleration effect in ℓ_2 attacks on CIFAR10 and ImageNet. Also, SAGE reduces 20%~35% queries in ℓ_∞ attacks.

To further demonstrate the acceleration process, we portray the distance drop curve as queries increase in Fig. 10. It is

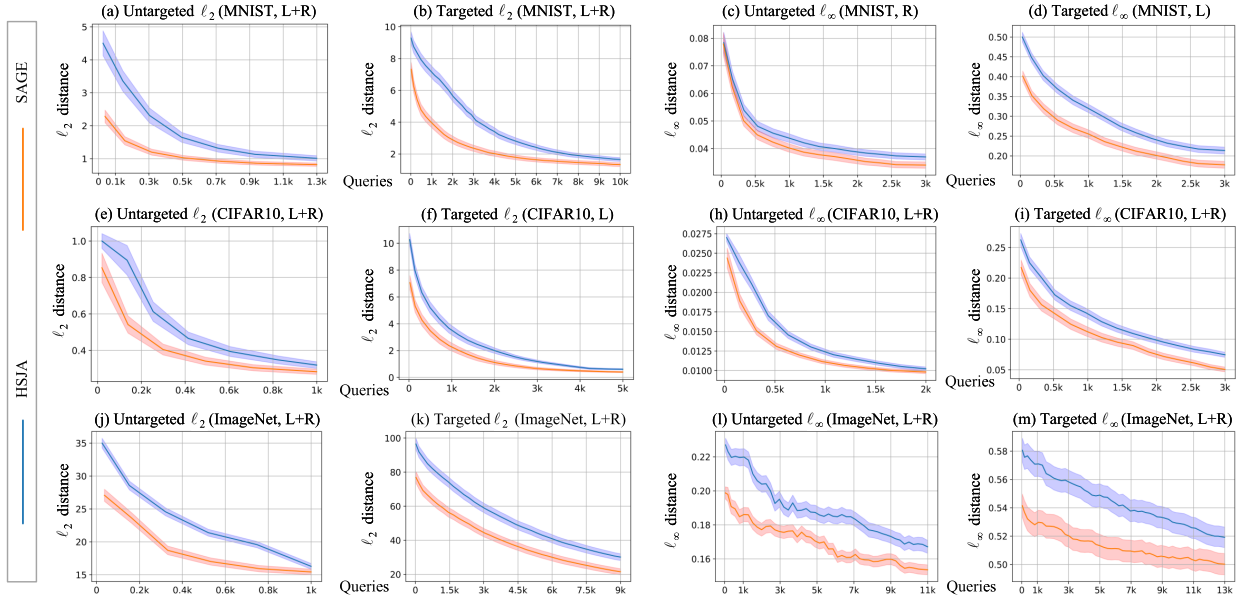


Fig. 10. Mean distance versus queries number on MNIST with LeNet, CIFAR10 with ResNet, and ImageNet with ResNet from top to bottom rows. 1st column: untargeted ℓ_2 . 2nd col.: targeted ℓ_2 . 3rd col.: untargeted ℓ_∞ . 4th col.: targeted ℓ_∞ .

clear that the distance in SAGE decreases faster than baseline, which means SAGE can get a high-quality AE with fewer queries. For different random seed settings, the standard deviations of SAGE and HSJA are almost the same. Sometimes, SAGE presents a slightly lower bias than HSJA, *e.g.*, shaded area in the subfigure (a)-(b). In addition, we observe some deep insights into SAGE. (i) With sLocator, SAGE can jump to a favorable position that reduces the initial distance before searching. For example, sLocator makes the initial ℓ_∞ distance decrease by about 0.1 in Fig. 10(d). (ii) With sRudder, the attack curve of SAGE has a larger gradient at the same distance (y-axis) thereby reaching success faster than HSJA, *e.g.*, Fig. 10(c). (iii) Combining sLocator and sRudder, SAGE can optimize the starting point, meanwhile accelerating the search process. For instance, in subfigure (k), SAGE finds an initial point with a smaller ℓ_2 distance, and then rapidly reduces the distance to generate successful AEs.

Remark 2: SAGE can reduce $\sim 20\%$ queries on decision-based attack. In SAGE, sLocator can jump to a favorable starting point which holds a small initial ℓ_p distance. While sRudder realizes guiding the perturbation to make the distance decreases faster.

D. Experiment 3: Acceleration on Score-Based Attack (RQ2)

In this section, we evaluate SAGE on two SOTA score-based attacks (NES [20] and AutoZOOM [10]). The definition of the successful attack is consistent with the baselines [41]: the thresholds of ℓ_∞ are 0.3, 0.05, and 0.05 for the MNIST, CIFAR10, and ImageNet, respectively. Meanwhile, the maximum number of queries for ImageNet is 50,000, and the other datasets are 4,000. To consider the more challenging scenario, we also test robust models (Madry's MNIST and CIFAR10 robust model [33]).

1) *Normal Model:* As Table II shows, we could summarize the attack results against the normal model as follows. SAGE

TABLE II
EVALUATE SAGE ON SCORE-BASED ATTACKS AGAINST THE NORMAL (TARGETED) AND ROBUST (UNTARGETED) MODELS. BASE, L, R, L+R RESPECTIVELY REPRESENT THE ORIGINAL METHOD, ACCELERATION WITH sLocator, WITH sRudder, AND WITH sLocator + sRudder

Model	Dataset	SAGE	AutoZOOM Success	Queries	NES Success	Queries
Normal (T)	MNIST	Base	91.3	1,471	77.5	2,544
		L	100.0	132	81.0	1,431
		R	91.5	966	79.8	1,397
		L+R	100.0	111	89.1	588
	CIFAR10	Base	92.9	1,117	98.8	1,078
		L	100.0	57	100.0	411
		R	100.0	479	100.0	563
		L+R	100.0	38	100.0	298
	ImageNet	Base	95.4	41,310	100.0	18,597
		L	100.0	19,696	100.0	8,476
		R	99.8	25,247	100.0	8,329
		L+R	100.0	12,354	100.0	7,261
Robust (U)	MNIST	Base	7.5	3,755	4.7	3,901
		L	94.8	221	88.1	476
		R	13.8	3,128	10.3	3,230
		L+R	94.9	216	92.8	288
	CIFAR10	Base	64.3	1,692	38.1	2,808
		L	100.0	11	98.4	314
		R	100.0	18	97.6	391
		L+R	100.0	9	98.0	271

achieve the 100% SUC on all three datasets. More importantly, SAGE reduce 70%~90% and 60%~80% queries for AutoZOOM and NES, respectively. Furthermore, we can find those three usages (sLocator, sRudder, sLocator + sRudder) can greatly improve SUC and reduce QUE. However, the combination of sLocator and sRudder works better than a single component. An interesting observation is that sLocator sometimes presents much better than sRudder. For instance,

the sLocator-only setting achieves QUE: 1,117 $\rightarrow$ 57 while the sRudder-only setting realizes QUE: 1,117 $\rightarrow$ 479 in CIFAR10 based on AutoZOOM. This could be because the background introduces a large ℓ_∞ distance, yet sLocator can effectively mitigate it.

2) *Robust Model*: When attacking the robust models which use adversarial training [33], SAGE can still achieve significant improvement. Note that these robust models use universal robust training rather than against specific attacks. In the bottom part of Table II, we find SAGE could reduce queries by 1~2 orders of magnitude. For instance, SAGE realizes QUE: 3,755 $\rightarrow$ 216, SUC: 7.5% $\rightarrow$ 94.9% for AutoZOOM on MNIST. In the three usages of SAGE, sLocator + sRudder still performs the best effect. On MNIST, the acceleration effect of sLocator is much better than sRudder. While there are slight differences between sLocator-only and sRudder-only on CIFAR10, about dozens of queries.

Remark 3: SAGE reduces over 60% queries for the score-based attack. The ensemble of sLocator and sRudder achieves the best results. Note that SAGE can significantly improve attack even against robust models that use adversarial training. This is because SAGE focuses on the salient features in pictures while does not rely on the model information.

E. Experiment 4: Compare With Hybrid Framework (RQ3)

We also compare the SOTA acceleration framework, the Hybrid attack [41], which combines score-based attack and transfer attack. Specifically, the Hybrid attack uses PGD [33] to determine an attack initial seed, then fine-tune local models with the generated AEs. At first, we use SAGE to accelerate the Hybrid attack. The results show that SAGE can further reduce queries for the Hybrid attack including no-tuned and tuned models. For instance, the Hybrid attack increases SUC from 91.3% to 98.9% for AutoZOOM on MNIST, and reduces the queries from 1,471 to 279 (Table II). SAGE further improves the Hybrid attack, SUC: 98.9% $\rightarrow$ 100%; QUE: 279 $\rightarrow$ 67 (Table III). To illustrate the advantages and disadvantages of two accelerators, we compare SAGE with the Hybrid attack in terms of acceleration effect and time overhead.

1) *Acceleration Effect*: We summarize the acceleration effect of SAGE and the Hybrid attack in Table IV. For the normal model, the optimization ratio of SAGE is roughly higher 5%~20% than the Hybrid attack on the MNIST and CIFAR10. On ImageNet, SAGE presents the better optimization effect, e.g., using ~40% fewer queries than the Hybrid attack.

For the robust model, the Hybrid attack holds a slight improvement (<5%). While SAGE can still fulfill >92% SUC and reduce the ~90% queries. Since the Hybrid attack needs to exploit PGD to initialize an attack starting seed and uses the generated AEs to fine-tune local models. It will be greatly limited when PGD fails against robust models. Moreover, it is not easy to fine-tune the local model in many real-world scenarios (such as Google Cloud API). Therefore, SAGE is more effective, flexible, and available than the Hybrid attack.

2) *Time Overhead*: To evaluate the time overhead, we randomly select 5,000 attack instances from ImageNet and

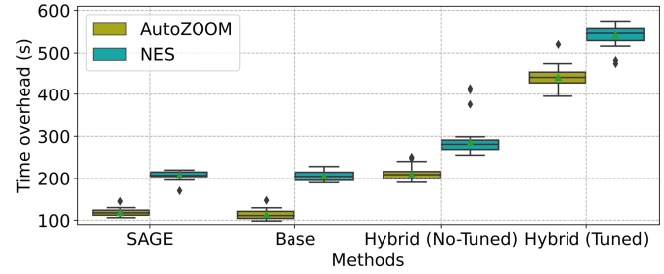


Fig. 11. Time overhead introduced by SAGE and the Hybrid framework.

calculate the average time introduced for 200 queries. Fig. 11 shows four groups of time overhead based on AutoZOOM and NES, including SAGE, baseline, no-tuned, and tuned approaches of the Hybrid attack. It is clear that SAGE introduces the negligible overhead, about 5s additional time cost. In contrast, the Hybrid attack will impose ~100s and ~300s for no-tuned and tuned methods. Note that SAGE requires fewer queries means less time. Therefore, our framework is more conducive to quickly implementing attacks in real scenarios.

Remark 4: For one thing, SAGE can further optimize the Hybrid attack. For another, SAGE fulfills the better acceleration effect, especially against robust models. Our proposed framework is more flexible since it does not rely on PGD or transfer attacks. Meanwhile, SAGE introduces the negligible time overhead, e.g., ~5s.

F. Experiment 5: Practical Evaluation (RQ4)

In this section, we perform practical evaluations on the Google Cloud Vision API [12] (GCV, for short) with unknown label sets. To attack GCV, it needs to cost \$0.001 per query. Particularly, we note that GCV has a very high number of near-redundant class labels. As shown in Fig. 12(a), the proposed labels include sports equipment, ball, and sports gear. Considering the relevance of those labels, we prefer to attack the sample into not similar classes.

Fig. 12 shows four sets of attack examples from ImageNet. Subfigure (a) is the original image that can be predicted as sports equipment. Fig. 12(b) and (c) show the AEs with $\ell_2 < 25.89$ from SAGE (228 queries, \$0.228) and HSJA (747 queries, \$0.747), respectively. Particularly, the AE's max-probability label is a cap in SAGE while it is a ball (similar to sports equipment) in HSJA. Fig. 12(d)-(f) show untargeted AEs with a limit of 1,000 queries (cost \$1). As shown in subfigure (e), the AE with $\ell_2 = 25.72$ from SAGE is identified as a carnivore and there is no dog in the predicted result. While the AE in 12(f) with $\ell_2 = 40.91$ obtained by HSJA is identified as a dog breed. It is still close to the dog.

Moreover, we also perform targeted attacks. Parts (h)-(j) describe a set of examples where the food is attacked as a bird. SAGE uses 10,006 queries to generate the AE in subfigure (i) yet HSJA requires 15,023 queries in subfigure (j). In other words, SAGE could save \$5.017 for this adversarial example compared to HSJA. Therefore, using SAGE to accelerate the attack has practical significance in real scenarios. For the

TABLE III
USING SAGE TO ACCELERATE THE HYBRID ATTACK AGAINST THE NORMAL (TARGETED) AND ROBUST (UNTARGETED) MODELS

Model	Dataset	SAGE	AutoZOOM-Hybrid Success	Queries	NES-Hybrid Success	Queries	AutoZOOM-Tuned Success	Queries	NES-Tuned Success	Queries
Normal (T)	MNIST	Base	98.9	279	89.2	892	99.5	194	92.2	671
		L	100.0	68	83.7	742	100.0	61	86.5	603
		R	99.1	237	89.0	806	99.6	163	90.2	615
		L+R	100.0	67	89.0	636	100.0	53	93.5	515
	CIFAR10	Base	98.2	271	99.8	339	96.3	459	99.6	427
		L	100.0	96	100.0	317	100.0	119	99.9	352
		R	100.0	124	100.0	167	100.0	62	99.8	365
		L+R	99.9	55	100.0	279	100.0	50	99.9	293
	ImageNet	Base	98.0	29,454	100.0	14,420	-	-	-	-
		L	100.0	14,747	100.0	8,182	-	-	-	-
		R	99.9	15,675	100.0	8,003	-	-	-	-
		L+R	100.0	9,536	100.0	7,191	-	-	-	-
Robust (U)	MNIST	Base	7.5	3,748	5.5	3,817	8.6	3,705	7.1	3,792
		L	95.4	197	92.9	290	94.8	217	93.2	245
		R	12.0	3,134	9.9	3,237	21.5	1,209	95.6	70
		L+R	95.2	204	93.0	280	100.0	12	92.9	256
	CIFAR10	Base	65.3	1,652	38.0	2,779	64.9	2,465	35.6	3,103
		L	100.0	9	96.4	365	100.0	9	94.6	366
		R	100.0	96	95.1	732	100.0	84	92.8	650
		L+R	100.0	23	97.1	331	100.0	63	93.7	392

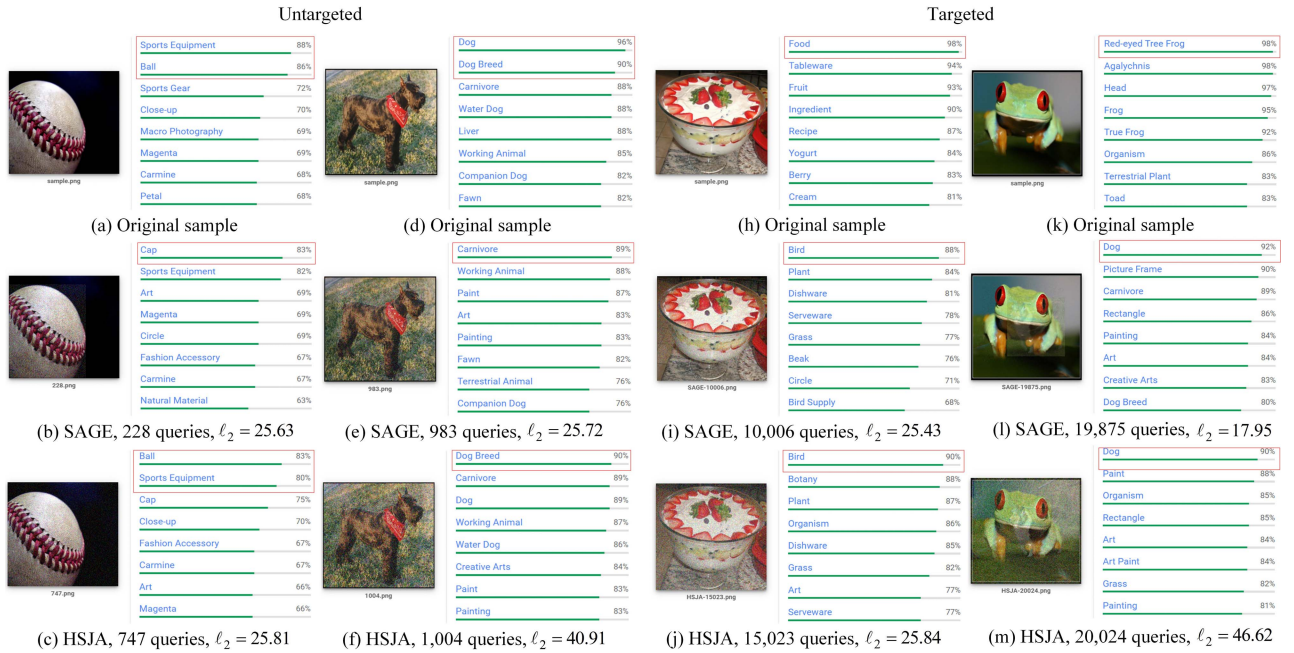


Fig. 12. Google Cloud Vision attack experiment. In this figure, (a)-(f) and (h)-(m) describe the untargeted attacks and the targeted attacks, respectively. The settings following, (a)-(c): same threshold $\ell_2 < 25.89$; (d)-(f): same queries (1,000); (h)-(j): same threshold $\ell_2 < 25.89$; (k)-(m): same queries (20,000).

last group samples, Fig. 12(k)-(m) attack the red-eyed tree frog to a dog, after 20,000 queries, SAGE generates AE with $\ell_2 = 17.95$, yet $\ell_2 = 46.62$ in HSJA.

G. Experiment 6: Evaluation on Different Image Selection (RQ5)

Considering the selected target images could determine the attack effect, we choose three types of instances (largest-distance, smallest-distance, and random) for comparison. In this experiment, we perform 5,000 targeted attacks on

ImageNet to satisfy $\ell_2 < 25.89$ and $\ell_\infty < 0.5$. The average queries are shown in Fig. 13, it is clear that choosing a close-distance target image can fastly acquire adversarial examples in both SAGE and HSJA. For instance, SAGE uses 7,916, 8,253, and 8,487 queries respectively when selecting the smallest, random, and largest initial image. While HSJA requires 10,186, 10,357, and 12,869 queries on average to fulfill successful attacks. Meanwhile, the query standard deviation in the random-target setting is more significant than in the smallest-distance and the largest-distance settings. Overall,

TABLE IV
COMPARE THE EFFECT OF SAGE WITH THE HYBRID FRAMEWORK

Model	Dataset	Method	Base	Query		Ratio(%)	
				Hybrid	SAGE	Hybrid	SAGE
Normal (T)	MNIST	AutoZOOM	1,471	194	111	86.81	92.45
		NES	2,544	671	588	73.62	76.89
	CIFAR10	AutoZOOM	1,117	271	38	75.74	96.60
		NES	1,078	339	298	68.55	72.36
	ImageNet	AutoZOOM	41,310	29,454	12,354	28.70	70.09
		NES	18,597	14,420	7,261	22.46	60.96
Robust (U)	MNIST	AutoZOOM	3,755	3,705	216	1.33	94.25
		NES	3,901	3,792	288	2.79	92.62
	CIFAR10	AutoZOOM	1,692	1,652	9	2.36	99.47
		NES	2,808	2,779	271	1.03	90.35



Fig. 13. Attack effect with different targeted images selection.

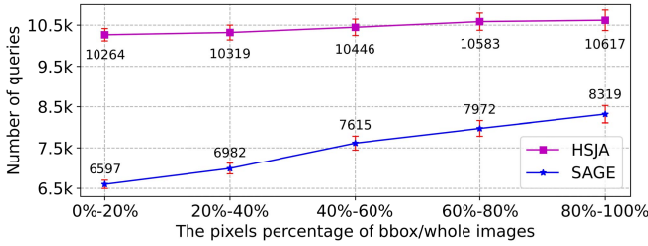


Fig. 14. Acceleration effect on diverse bbox sizes.

SAGE can indeed accelerate the attack no matter which initial image is selected.

Remark 5: SAGE can achieve the significant acceleration effect regardless of which target sample is selected.

H. Experiment 7: Evaluation on Diverse Bbox (RQ6)

In this section, we evaluate the acceleration effect with different bbox sizes and numbers.

1) *Bbox Size*: To evaluate SAGE on different bbox sizes, we divide 5,000 ImageNet samples into 5 intervals. Specifically, supposing the pixels contained in the bbox as P_{bbox} , and the pixels contained in the image as P_{all} , then the bbox proportion pro_{bbox} can be calculated as $pro_{bbox} = P_{bbox} / P_{all}$. The five intervals based on pro_{bbox} are: [0%, 20%], (20%, 40%], (40%, 60%], (60%, 80%], (80%, 100%).

As Fig. 14 shows, the acceleration effect of SAGE is more significant when pro_{bbox} is small. Particularly, SAGE can reduce $\sim 40\%$ queries when $pro_{bbox} \in [0\%, 20\%]$. This is because SAGE can perturb accurately those critical pixels that contribute more to the prediction result. Even if the bbox occupies $> 80\%$ pixels, SAGE could still maintain a good acceleration effect, e.g., QUE: 10,617 $\rightarrow$ 8,319. Yet HSJA has only slight changes for different bbox sizes since its attacks area are distributed on the whole picture. Furthermore, as the

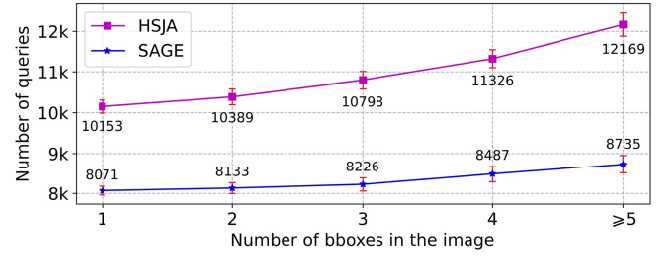


Fig. 15. Acceleration effect on different bbox numbers.

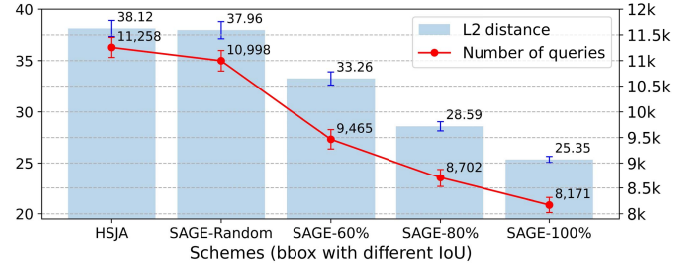


Fig. 16. Evaluate SAGE on imprecise bbox. (i) The bar graph shows the ℓ_2 distance after 8K queries; (ii) The line chart shows required queries when $\ell_2 < 25.89$.

pro_{bbox} increases, the standard deviation of queries tends to be larger.

2) *Bbox Number*: If the original image contains multiple objects, the difficulty of attack will increase because the adversary needs to perturb each object. Thus, we test the performance of SAGE for different bbox numbers. We divide the 5,000 ImageNet instances according to the bboxes number, i.e., {[1], [2], [3], [4], and [5, ∞]}. The attack results of SAGE and HSJA are summarized in Fig. 15. As the bbox number increases, the required queries increase in both two attacks. Nonetheless, SAGE can reduce 20%~40% queries on HSJA in various situations. Moreover, as the number of bboxes increases, the standard deviation of query number tends to be higher. Overall, SAGE could fulfill good acceleration in a stable manner regardless of the size and number of bboxes. Especially the concentrated objects in images will support SAGE to realize more efficient attacks.

Remark 6: SAGE can realize a good acceleration effect in a stable manner with the different number of bboxes or different bbox sizes.

I. Experiment 8: Evaluation on Imprecise Bbox (RQ7)

Our core idea is exploiting the bbox to capture salient features, so we consider evaluating SAGE with an imprecise bbox. In this experiment, we manually offset the ground-truth labels to get inaccurate bboxes with 60% and 80% Intersection-over-Union (IoU) [35]. Note that the typical object detection methods could generally achieve $> 60\%$ IoU [35]. We randomly selected 5,000 ImageNet samples to perform the ℓ_2 targeted attacks in five situations including SAGE (100%-IoU bbox), SAGE (80%-IoU bbox), SAGE (60%-IoU bbox), SAGE with random segmentation bbox, and HSJA, respectively. In Fig. 16, we calculate the results

containing two parts: (i) the ℓ_2 distance under 8K queries; (ii) the required QUE when the $\ell_2 < 25.89$. We find that SAGE presents a remarkable acceleration effect in all 60%, 80%, and 100% IoU, while only a slight change between SAGE on random bbox and baseline. Meanwhile, the more accurate bbox, the more efficient and stable SAGE is. For example, to satisfy $\ell_2 < 25.89$, SAGE needs 8,171, 8,702, and 9,465 queries on 100%, 80%, and 60% IoU bbox respectively. Even if on the 60%-IoU bbox, SAGE still has a good acceleration effect, *i.e.*, ℓ_2 distance (under 8K queries): $33.26 < 38.12$; QUE (when $\ell_2 < 25.89$): $9,465 < 11,258$.

This phenomenon can be attributed to the expansion mechanism in sLocator (§ V-B). When the bbox holds an accuracy loss that could not contain most of the salient features, the starting image will not be predicted as the target class. Then the expansion mechanism is triggered, thereby the neglected salient pixels due to the imprecise bbox will gradually be considered. Therefore, the originally imprecise bbox can still contain most of the salient features after expansion. It is interesting that as the field of computer vision advanced, object detection and semantic segmentation technologies will become more accurate, which could make SAGE attacking more concentrated on salient features. We hope that our proposed framework in this paper can provide a new perspective on adversarial example generation and benefit the subsequent attack and defense research.

Remark 7: With the more accurate bbox, SAGE tends to be more effective. Even using the imprecise bbox (e.g., 60% IoU), SAGE still realizes the good acceleration effect.

VII. DISCUSSION

A. Visualization of Attack Process

To observe and analyze the attack process for SAGE and baseline, we use t-SNE [32] to visualize the adversarial example generation. Fig. 17 shows a targeted attack instance that SAGE uses 2,830 to obtain the final AE, and HSJA uses 6,036 queries. During the attack of SAGE, the sample can approach the target explicitly and quickly in subfigure (a). While the perturbation of HSJA is more chaotic and curved, and also presents many fluctuations orthogonal to the target direction. Those chaotic and curved perturbations may be the main reason for the inefficiency of black-box attacks. Yet SAGE optimizes the attack process and reduces the perturbations orthogonal to the target direction.

B. Extensibility of SAGE

In different application domains, we can apply the core idea of SAGE to optimize the attack. For instance, consider attacking a malware classifier. We can analyze static feature vectors (*e.g.*, activities and permissions) and depict a series of fingerprints for each category. This list of fingerprints can be considered as a bbox set. When performing an adversarial attack, extract the fingerprint presented in the input sample and concentratedly tamper the information on its fingerprint area. Essentially, SAGE aims to determine the critical area for model inference and modify it.

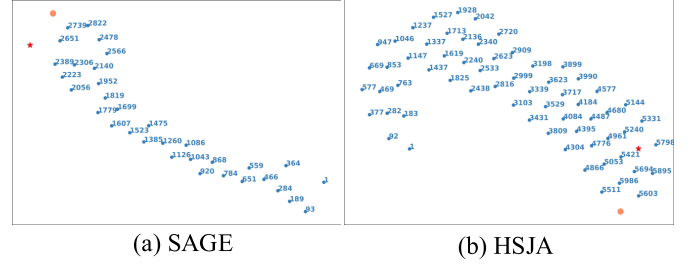


Fig. 17. Visualize the attack process for SAGE and HSJA. In the figure, the number '1', orange circle, and red star represent the initial point of the attack, the original image, and the final AE, respectively. Moreover, the label of each point is the number of queries. Note that the distance in the figure is not equal to the ℓ_p distance due to the projection transformation.

C. Defenses for SAGE

Consider the core idea of SAGE, which finds the significant area which determines the model prediction results and then perturbs it. While current DNN classification model designs often refer to the following approach. They perceive multiple feature areas in a context-aware manner through the processing of a feature pyramid. Then the model aggregates multiple feature areas and determines possible predicted classification results. This architecture focuses on the semantic information between adjacent pixels. Therefore, we propose to design a DNN model with feature-aware decentralization, which tends to reduce the dependence on regional pixels in a robust manner. For instance, Ribeiro et al. [39] mention a classifier could differentiate wolves from huskies based on the image background, given all training-set pictures of wolves had snow in the background while pictures of huskies did not. This phenomenon is that the model picks up undesirable correlations, but it shows a potential possibility that model inference can depend on more than the foreground area. In other words, it could be a possible defense strategy that makes model prediction results depending on the many discrete pixels distributed on the whole image.

Moreover, a series of studies [21], [23] have shown that the existence of adversarial examples could be attributed to the model learning the non-robust features. As some leading work, Sicheng et al. [51] present using the worst-case mutual information to capture the robust representations. And Shivam et al. [14] analyze the robust features from a spectral view. Specifically, if the output is insensitive to input perturbation, these features are considered robust. We will explore it in future work.

VIII. CONCLUSION

In this paper, we propose an acceleration framework SAGE to reduce the queries and improve the success rate for black-box attacks. SAGE contains two components: sLocator and sRudder, which can optimize the attack starting point and the search process, respectively. We first formulate that the saliency map can accelerate the AE generation by guiding the perturbations toward the most adversarial direction. Then we advance using the bounding box to capture the salient pixels, which overcomes the problem of gradient information

lack in the black-box attack. Meanwhile, we provide a series of observations that demonstrate bbox holds *model invariance* and *process invariance*. Based on the above insights, we carefully design the acceleration framework SAGE and perform extensive evaluations. In our experiments, SAGE could reduce >20% queries and improve the attack success rate to 95%~100%. Especially SAGE presents the fundamental improvement against robust models that use adversarial training. Compared with the Hybrid attack framework, SAGE is more flexible and efficient. Moreover, SAGE can be applied in real-world scenarios, *e.g.*, attack the Google Cloud Vision API. We finally visualize the AE generation to provide deep insights for SAGE and the existing black-box attack, as well as suggest possible defenses to be investigated in future work.

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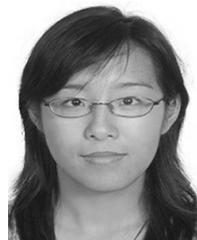
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